Week 4 | Lab Practice 04

Software Project Management: Feasibility, Scope, SMART Objectives and WBS
Instructor: Dr. Hussain | Zhengzhou University | School of Computer and Artificial Intelligence
Duration: 1 hour 30 minutes (90 minutes) | Practice only - no marks

What you completed in the previous lab

You already have an Overleaf project proposal. In Lab Practice 03, you added Software Requirements, System Design, and Project Estimation sections; wrote at least five functional and three non-functional requirements; created Use Case, Activity, and Class UML diagrams; inserted figures and a numbered equation; and added citations using a bibliography file.

Important: Continue in the SAME Overleaf project and the SAME main .tex file. Do not create a new Overleaf project, remove previous sections, or replace your existing UML diagrams. All new examples must be adapted to your own selected software project.

90-minute lab schedule

Time	Activity
0-5 min	Open and check the existing Overleaf proposal
5-25 min	TELOS feasibility analysis
25-35 min	Economic feasibility: ROI and simple payback
35-50 min	Scope statement: inclusions and exclusions
50-60 min	Three SMART objectives
60-80 min	Create and insert a WBS diagram
80-85 min	Brief WBS Dictionary (three work packages)
85-90 min	Compile, verify, and submit

Step 1. Open your existing Overleaf project (5 minutes)

1. Sign in to Overleaf and open the SAME proposal used in Lab Practices 02 and 03. 2. Confirm your requirements, UML figures, equation, and references still compile. 3. After the existing Project Estimation section, add these new sections:

```
\section{Feasibility Study}
\section{Project Scope}
\section{SMART Objectives}
\section{Work Breakdown Structure}
\section{WBS Dictionary}
```

Expected output: Five new section headings in the same proposal.

Step 2. Conduct a TELOS feasibility study (20 minutes)

Under Feasibility Study, add a subsection for each dimension. For YOUR project, write two sentences per dimension: one describing the issue and one giving evidence or a clearly labeled assumption.

```
\subsection{Technical Feasibility}
\subsection{Economic Feasibility}
\subsection{Legal Feasibility}
\subsection{Operational Feasibility}
\subsection{Schedule Feasibility}
```

Answer: Technical - Do we have the skills, tools, and integrations? Economic - Are expected benefits worth the costs? Legal - Are privacy and licensing requirements satisfied? Operational - Will users adopt the system? Schedule - Can the team meet its deadline? End with a decision: **GO, MODIFY, or STOP**, and justify it.

Step 3. Calculate economic feasibility (10 minutes)

Use illustrative values for your own project. Example: initial cost = CNY 20,000; annual net benefit = CNY 10,000; evaluation period = 3 years; assume no other costs for this simplified exercise.

```
\subsection{ROI and Payback Period}
\begin{equation}
ROI = \frac{30000 - 20000}{20000} \times 100 = 50\%
\end{equation}
\begin{equation}
Payback = \frac{20000}{10000} = 2 \text{ years}
\end{equation}
```

Replace these example numbers with your own **labeled assumptions**. Explain what the results suggest. This is a simplified, undiscounted calculation, not an NPV analysis.

Step 4. Define project scope (15 minutes)

Use your existing functional requirements and UML Use Case Diagram. Identify at least four features that are **in scope**, at least two that are **out of scope**, and one assumption or constraint. Make sure the scope agrees with the requirements you wrote last week.

```
\subsection{In Scope}
\begin{itemize}
  \item [Replace with approved feature 1]
  \item [Replace with approved feature 2]
\end{itemize}
\subsection{Out of Scope}
\begin{itemize}
  \item [Replace with excluded feature 1]
\end{itemize}
```

Expected output: A clear boundary statement for your own system, not the sample library app.

Step 5. Write three SMART objectives (10 minutes)

Write three objectives for your selected system. Each must specify a deliverable or improvement, a measurable target, and a deadline. Add one short sentence explaining why it is achievable and relevant.

Example only: "By Week 6, complete the task-creation feature with at least 95% of defined functional tests passing." Adapt it to your project. Avoid vague words such as "fast" or "secure" without testable criteria.

Step 6. Draw and insert your WBS (20 minutes)

1. Open diagrams.net. 2. Put your project name at the top (1.0). 3. Identify at least three major deliverables (for example, 1.1, 1.2, 1.3). 4. Break them into smaller outputs, using WBS codes such as 1.1.1. 5. Check the 100% Rule: all approved project work must be covered, with no duplicated scope. 6. Export the diagram as **wbs.png** and upload it to the SAME Overleaf project.

```
\subsection{WBS Diagram}
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.9\textwidth]{wbs.png}
  \caption{Work Breakdown Structure for Our Project}
  \label{fig:wbs}
\end{figure}
Figure~\ref{fig:wbs} shows our project deliverables.
```

If your preamble does not already contain it, add **\usepackage{graphicx}** before \begin{document}.

Step 7. Create a short WBS Dictionary (5 minutes)

Choose three lowest-level work packages from your diagram. For each, provide: WBS code, deliverable name, estimated effort in hours, and one testable acceptance criterion. Use a simple table in Overleaf. These are preliminary estimates that can be improved in the next estimation lab.

Step 8. Final check and submission (5 minutes)

Recompile the SAME Overleaf proposal. Check that old requirements and UML diagrams remain, new sections appear, WBS image is visible, equations display correctly, and references still work. Submit the updated PDF, main .tex file, .bib file, and new wbs.png along with the existing diagram images, following your class submission method.

Reminder: This is a hands-on practice session with **no marks**. Keep the same project throughout the semester so you can use this WBS in future estimation and scheduling labs.